

Determination of charge to mass ratio of electron

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Abstract— This experiment was aimed to determine the charge-to-mass ratio of the electron with a Thomson tube. The ratio was calculated from the relationship between the deflection voltage and magnetic field when null deflection was observed. Linear regression in python was used to evaluate gradient from current-voltage graph to obtain e/m value of $1.78 * 10^{11} \pm 0.03 * 10^{11} \text{ Ckg}^{-1}$ which was consistent with theoretical expectations, demonstrating the null-deflection method was appropriate.

I. INTRODUCTION WITH THEORY

This experiment aims to obtain the charge-to-mass ratio ($\frac{e}{m}$) of electron by achieving null deflection from corresponding J.J. Thomson first measured $\frac{e}{m}$ of the electron by studying the behavior of charged particles in electric and magnetic fields [1]. The underlying principles rely on the forces experienced by an electron in uniform electric ($\mathbf{E}$) and magnetic ($\mathbf{B}$) fields:

$$F_E = eE \quad (1)$$

$$F_B = evB \quad (2)$$

Where:

- F_E is the force on charged particle
- e is charge of electron
- E is the electric field strength
- v is the velocity of electron
- B is the magnetic field strength

When the two fields are applied orthogonally and balanced, as show in Fig. 1, the electron travels undeflected, leading to:

$$eE = evB \quad (3)$$

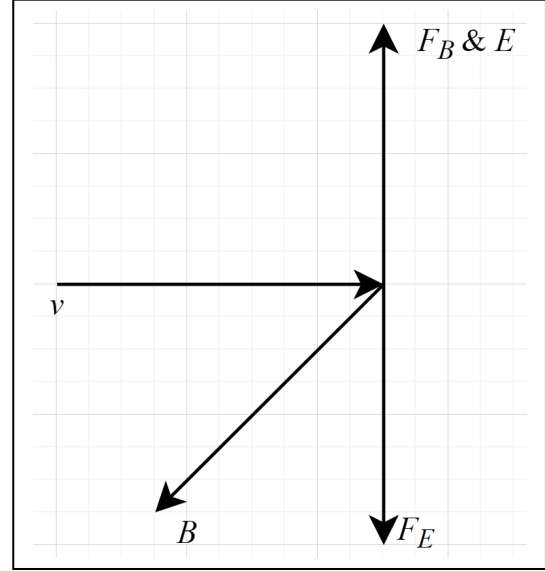


Fig. 1 Illustration of orthogonally balanced electric and magnetic field forces applied on the electron, where all line segments are perpendicular to each other.

Therefore:

$$v = \frac{E}{B} \quad (4)$$

Electrons are accelerated through the electron gun voltage (V_a), converting electrical potential energy into kinetic energy in principle of conservation of energy:

$$eV_a = \frac{1}{2}mv^2 \quad (5)$$

Where m is the mass of the particle.

Substituting for v yields the charge-to-mass ratio:

$$\frac{e}{m} = \frac{2V_a}{B^2d^2} \quad (6)$$

where d is the separation between deflector plates.

With known applied electric and magnetic field strengths it is possible to determine the $\frac{e}{m}$ value of the electron, which is also the aim of the experiment [2].

In which, the uniform magnetic field strength is given by:

$$B = kI_H \quad (7)$$

Where I_H is the current through the Helmholtz coils and $k = 4.17 * 10^{-3} \text{ TA}^{-1}$. [2]

The electric field strength is, on the other hand, is given by:

$$E = \frac{V_p}{d} \quad (8)$$

II. DESCRIPTION OF EQUIPMENT

- TEL 2502 HELMH027Z Thomson Tube containing an electron gun and Helmholtz coils generating homogenous magnetic and electric fields.

- AimTTi EL302P-USB Programmable PSU (30V, 2A) converting AC current into DC current for use in the coil circuit and altering current through the Helmholtz coils (I_H).
- 2 * TENMA 72-7925 Multi-meter for measuring V_a and plate voltage (V_p)
- V_p power supply
- $\frac{1}{1000}$ potential divider to reduce V_a to a level that it can be safely measured
- Griffin Voltline 5kV EMT Supply for V_a

III. PROCEDURE

The apparatus was set up as per Figure 2 below. V_a was set to 2.98 kV in and V_p was set to start from 53.7 V in experiment 1, in the repeated experiment 2 with a different PSU the values are 3.05 kV and 51.8 V, respectively. A deflected beam of blue light was observed then the I_H is altered to nullify the beam's deflection. The I_H value at null deflection is recorded.

The previous step was repeated for increased V_p at increments of approximately 25 V.

A plot of $I_H - V_p$ was created with its gradient determined, which was then used to calculate the $\frac{e}{m}$ value using the derived formula:

$$\frac{e}{m} = \frac{\text{Gradient}^2 * V_a}{4.49 * 10^8} \quad (9)$$

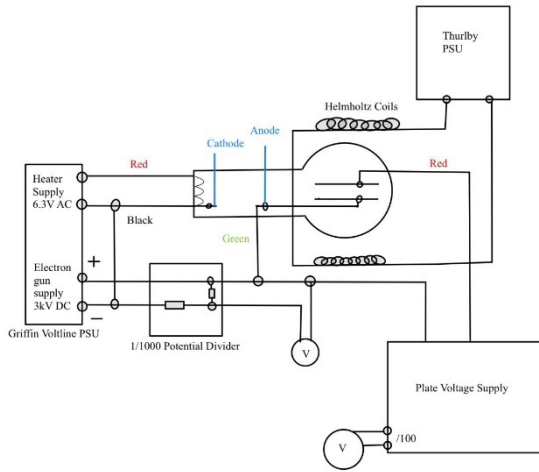


Figure 2. Illustration of the experiment setup

IV. RESULT AND ANALYSES

Experiment 1:

Index	V_p/V	I_H/A	$\Delta V_p/V$	$\Delta I_H/A$
1	53.7	0.060	1	0.005
2	75.0	0.070	1	0.005
3	100	0.100	1	0.005
4	131	0.125	1	0.005
5	151	0.140	1	0.005
6	178	0.165	1	0.005
7	203	0.190	1	0.005
8	227	0.210	1	0.005
9	251	0.230	1	0.005
10	275	0.250	1	0.005
11	299	0.270	1	0.005

Table 1. Results from Experiment 1.

The selection of ΔV_p was based on the precision of the multi-meter, and ΔI_H was based on the precision of the PSU and estimation if null deflection is not achieved on consecutive I_H values.

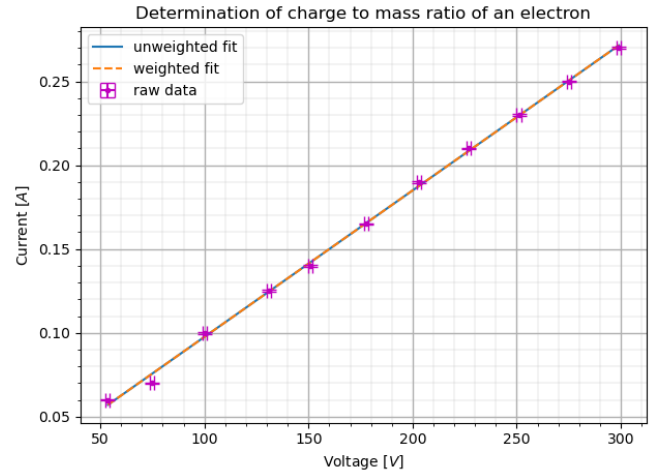


Figure 3. $I_H - V_p$ diagram of Experiment 1.

From the graph, the $\frac{e}{m}$ value of electron is obtained to be $1.97 * 10^{11} \pm 2.31 * 10^9 \text{ Ckg}^{-1}$, a python program was used to calculate the values and propagate the error involved.

The value is not consistent with the cited value of $1.76 * 10^{11} \text{ Ckg}^{-1}$ [3] with an error of 12 %.

It is found that the PSU used in experiment 1 had a display of I_H with the precision of 2 decimal places and could only adjust the value in increments of 0.01A, while other PSUs provided had a display with the precision of 3 decimal places with stepless adjustments.

Since the null deflection to record the I_H value is mostly determined by raw eye, the error would be significant. A repeated experiment 2 using a different model of PSU was conducted to eliminate this effect.

Experiment 2:

Index	V_p/V	I_H/A	$\Delta V_p/V$	$\Delta I_H/A$
1	51.8	0.041	1	0.001
2	76.0	0.061	1	0.001
3	97.5	0.083	1	0.001
4	123	0.125	1	0.001
5	150	0.140	1	0.001
6	176	0.165	1	0.001
7	202	0.190	1	0.001
8	225	0.210	1	0.001
9	248	0.230	1	0.001
10	276	0.250	1	0.001
11	298	0.270	1	0.001

Table 2. Results from repeated Experiment 2.

Both selections of uncertainties are based on the precision of the measuring apparatus used.

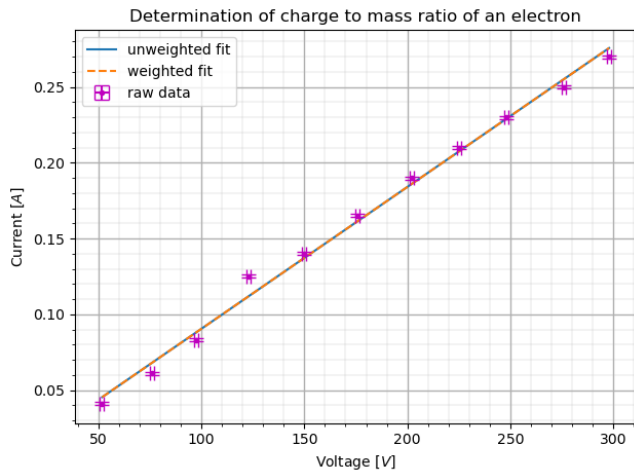


Figure 4. $I_H - V_p$ diagram of Experiment 2.

Using the same methods as in Experiment 1, value of electron is obtained to be $1.78 \times 10^{11} \pm 0.03 \times 10^{11} \text{ Ckg}^{-1}$. Chi-squared test was also conducted using the python program, which gave a reduced χ^2 value of 0.385 indicating that linear regression is an overfitting. The χ^2 values are the same for weighted and unweighted fits since they are identical due to the systematic error was constant as shown in the data tables [4]. Nevertheless, considering that the residuals are small, linear fit would still be appropriate, human and random error would be a probable cause for this deviation.

V. CONCLUSION

Comparing the obtained value from Experiment 2 with the cited value, the percentage error is only 1.1 %. The minor discrepancies could be attributed to non-uniform magnetic fields at the edges of the Helmholtz coils and potential misalignment in the electron gun, as well as lack of accuracy from the measuring apparatus. The accuracy could be improved by calibrating the Helmholtz coil placement more precisely, and using more accurate measuring apparatus. Further work could consist of repeating with alternative methods, to corroborate results.

REFERENCES

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